

Local-field excitations in 2D lattices of resonant atoms

S. N. Volkov and A. E. Kaplan

*Dept. of Electrical and Computer Engineering,
Johns Hopkins University, Baltimore, MD 21218*

(Dated: February 12, 2010)

Abstract

We study excitations of the local field (locsitons) in nanoscale two-dimensional (2D) lattices of strongly interacting resonant atoms and various unusual effects associated with them. Locsitons in low-dimensional systems and the resulting spatial strata and more complex patterns on a scale of just a few atoms were predicted by us earlier [A. E. Kaplan and S. N. Volkov, *Phys. Rev. Lett.* **101**, 133902 (2008)]. These effects present a radical departure from the classical Lorentz-Lorenz theory of the local field (LF), which assumes that the LF is virtually uniform on this scale. We demonstrate that the strata and patterns in the 2D lattices may be described as an interference of plane-wave locsitons, build an analytic model for such unbounded locsitons, and derive and analyze dispersion relations for the locsitons in an equilateral triangular lattice. We draw useful analogies between one-dimensional and 2D locsitons, but also show that the 2D case enables locsitons with the most diverse and unusual properties. Using the nearest-neighbor approximation, we find the locsiton frequency band for different mutual orientations of the lattice and the incident field. We demonstrate a formation of distinct vector locsiton patterns consisting of multiple vortices in the LF distribution and suggest a way to design finite 2D lattices that exhibit such patterns at certain frequencies. We illustrate the role of lattice defects in supporting localized locsitons and also demonstrate the existence of “magic shapes”, for which the LF suppression at the exact atomic resonance is cancelled.

PACS numbers: 42.65.Pc, 85.50.-n